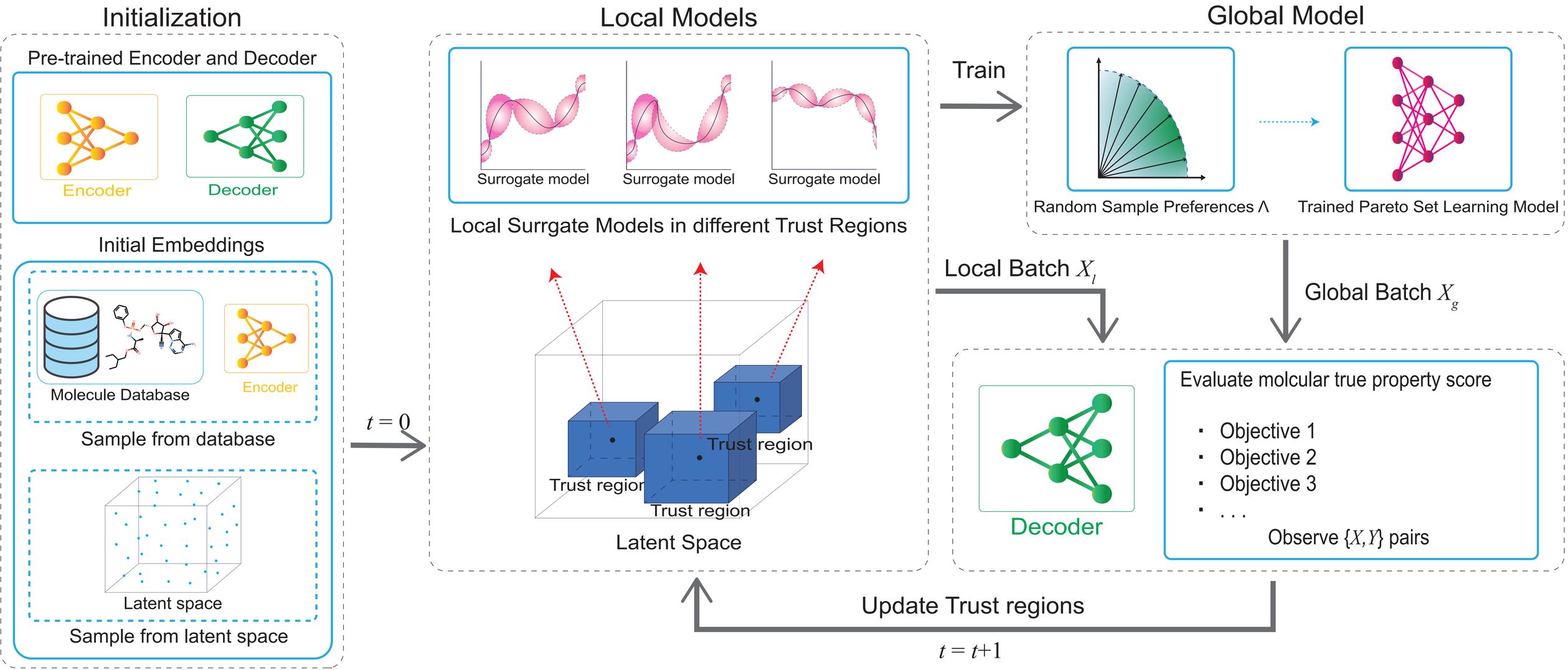
Initialization
Pre-trained Encoder and Decoder
Encoder
Decoder
Initial Embeddings
Molecule Database
Encoder
Sample from database
Latent space
Sample from latent space
$t = 0$
Local Models
Surrogate model
Surrogate model
Surrogate model
Local Surrgate Models in different Trust Regions
Trust region
Trust region
Trust region
Latent Space
Train
Global Model
Random Sample Preferences Λ
Trained Pareto Set Learning Model
Local Batch X_l
Global Batch X_g
Decoder
Evaluate molcular true property score
Objective 1
Objective 2
Objective 3
. . .
Observe $\{X, Y\}$ pairs
Update Trust regions
$t = t+1$